

CURIONEST

CHEMISTRY MEASUREMENT

Lab Tools, Density & Percent Error

Visual worksheet for U.S. Grades 8-10

STUDENTS WILL

- identify common measurement tools
- justify equipment choices with visual evidence
- calculate density and percent error
- compare accuracy and precision
- repair a measurement misconception



4 student pages
4-page full key
Editable DOCX

PRINT-READY • US LETTER

$$\text{density} = \text{mass} / \text{volume}$$

TEACHER QUICK START

Chemistry Measurement Lab Tools & Density Visual Worksheet

AT A GLANCE

U.S. Grades 8-10 | 35-45 minutes | Individual, pairs, or intervention | No lab materials required

Purpose	Students connect visible equipment features to measurement choices, then apply the resulting data to density, percent error, accuracy, and precision.
Suggested pacing	5 min quick reference; 10 min tool selection; 15 min calculations; 10 min lab case and repair; 5 min review.
Before class	Print pages 1-4. Decide whether students may use calculators. Review local expectations for significant figures and liquid-volume reading.
Differentiation	Support: allow the formula toolbox and a unit bank. Core: complete as written. Extension: ask students to design one additional measurement trial that would improve confidence in precision.
Look-fors	Students name the measured quantity, not only the tool; carry units through calculations; use the accepted value in the percent-error denominator; avoid claiming precision from a single trial.
Safety note	This is a paper-based readiness task. Local laboratory rules, teacher directions, SDS information, and equipment-specific procedures always take priority.

Fast formative check

Ask: "What makes a measurement trustworthy?" A complete response should connect the right tool, readable scale, unit, repeated trials, and transparent calculation.

Measurement Ready

Use the tool, scale, unit, and calculation as one evidence chain.

Name: _____ Date: _____ Class Period: _____

LEARNING TARGETS

Match common lab tools to the quantity and unit they measure. • Choose an appropriate tool and justify the choice with visual evidence. • Calculate density and percent error with correct units and reasonable precision.

Three tools, three different jobs



Electronic balance

Measures mass
Typical unit: g



Graduated cylinder

Measures liquid volume
Typical unit: mL



Beaker

Holds and mixes
Volume is approximate

Measurement rules that prevent avoidable errors

- 1 Record the value and the unit together.
- 2 For an analog scale, record all certain digits plus one estimated digit.
- 3 Read a concave liquid meniscus at eye level from its lowest point.
- 4 Keep units through every calculation and round only the final result.

FORMULA TOOLBOX

density = mass / volume

percent error =
|experimental - accepted| / accepted × 100

Units tell the story

mass: g
liquid volume: mL
density: g/mL

Choose the Tool

Name what you see, then use visible features to justify the choice.

Name: _____ Date: _____ Class Period: _____

A. Identify each instrument

1



Instrument: _____

Quantity/job: _____

Unit (if measured): _____

2



Instrument: _____

Quantity/job: _____

Unit (if measured): _____

3



Instrument: _____

Quantity/job: _____

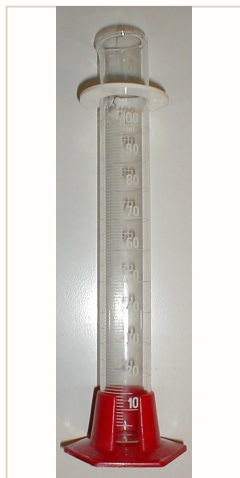
Unit (if measured): _____

B. Select and justify

4. You need 18.6 mL of water. Choose a beaker or a graduated cylinder. Explain which visible design features support your choice.

5. You need about 125 mL of solution and room to stir. Choose a beaker or a graduated cylinder. Explain.

C. Read visual evidence



6. Identify two visible features that show this vessel was designed to measure liquid volume.

Photo: Darrien / Haltopub, Public Domain, Wikimedia Commons



Calculate with Units

Show the formula, substitute values with units, and round only the final answer.

Name: _____ Date: _____ Class Period: _____

GUIDED EXAMPLE

A sample has a mass of 36.4 g and a volume of 14.0 mL.

$$d = m / V = 36.4 \text{ g} / 14.0 \text{ mL} = 2.60 \text{ g/mL}$$

7. A liquid sample has a mass of 27.0 g and a density of 2.70 g/mL. Calculate its volume.

Work space

8. A liquid has a density of 0.789 g/mL and a volume of 25.0 mL. Calculate its mass.

Work space

9. A student measures 2.58 g/mL. The accepted value is 2.70 g/mL. Calculate percent error.

Work space

10. Accuracy and precision

Accepted density: 2.70 g/mL

Set A	2.70, 2.71, 2.69 g/mL
Set B	2.42, 2.41, 2.42 g/mL

Which set is accurate? Which is precise? Explain using the pattern in the data.



Reason from Evidence

Use the data to build a claim, diagnose the error, and plan a repair.

Name: _____ Date: _____ Class Period: _____

LAB CASE: UNKNOWN LIQUID

A student uses an electronic balance and a graduated cylinder. The empty container has a mass of 42.31 g. The container plus liquid has a mass of 67.19 g. The liquid volume is 20.0 mL. The accepted density is 1.25 g/mL.

11. Mass of liquid: _____

12. Experimental density: _____

13. Percent error: _____

14. Claim - Evidence - Reasoning

Claim: Is the experimental result close to the accepted value?

Evidence: Cite at least two numbers from the case.

Reasoning: Explain what percent error tells you and what it does not tell you about precision.

15. Error diagnosis

Another student reads the liquid above eye level, rounds every intermediate value, and omits units. Name two problems and one exact repair action.

REPAIR LOG

Item/skill to repair: _____

Next action: _____

Incorrect idea or step: _____

When I will retry: _____

1

Measurement Ready

Use the tool, scale, unit, and calculation as one evidence chain.

LEARNING TARGETS

Match common lab tools to the quantity and unit they measure. • Choose an appropriate tool and justify the choice with visual evidence. • Calculate density and percent error with correct units and reasonable precision.

Three tools, three different jobs



Electronic balance

Measures mass
Typical unit: g



Graduated cylinder

Measures liquid volume
Typical unit: mL



Beaker

Holds and mixes
Volume is approximate

Measurement rules that prevent avoidable errors

- 1 Record the value and the unit together.
- 2 For an analog scale, record all certain digits plus one estimated digit.
- 3 Read a concave liquid meniscus at eye level from its lowest point.
- 4 Keep units through every calculation and round only the final result.

FORMULA TOOLBOX

density = mass / volume

percent error =
|experimental - accepted| / accepted × 100

Units tell the story

mass: g
liquid volume: mL
density: g/mL

2

Choose the Tool

Name what you see, then use visible features to justify the choice.

A. Identify each instrument

1



Electronic balance
mass
g

2



Graduated cylinder
liquid volume
mL

3



Beaker
holding/mixing; rough volume only

B. Select and justify

4. You need 18.6 mL of water. Choose a beaker or a graduated cylinder. Explain which visible design features support your choice.

Graduated cylinder. Its narrow shape and graduated scale support a more precise volume reading than a beaker.

5. You need about 125 mL of solution and room to stir. Choose a beaker or a graduated cylinder. Explain.

Beaker. The task asks for an approximate amount and mixing space, not a precise delivered volume.

C. Read visual evidence



6. Identify two visible features that show this vessel was designed to measure liquid volume.

Any two visible features: tall/narrow vessel, graduated marks, numbered scale, stable base, or pouring lip.

3

Calculate with Units

Show the formula, substitute values with units, and round only the final answer.

GUIDED EXAMPLE

A sample has a mass of 36.4 g and a volume of 14.0 mL.

$$d = m / V = 36.4 \text{ g} / 14.0 \text{ mL} = 2.60 \text{ g/mL}$$

7. A liquid sample has a mass of 27.0 g and a density of 2.70 g/mL. Calculate its volume.

10.0 mL

8. A liquid has a density of 0.789 g/mL and a volume of 25.0 mL. Calculate its mass.

19.7 g

9. A student measures 2.58 g/mL. The accepted value is 2.70 g/mL. Calculate percent error.

4.44%

10. Accuracy and precision

Accepted density: 2.70 g/mL

Set A 2.70, 2.71, 2.69 g/mL

Set B 2.42, 2.41, 2.42 g/mL

Which set is accurate? Which is precise? Explain using the pattern in the data.

Set A is accurate and precise. Set B is precise but not accurate.

4

Reason from Evidence

Use the data to build a claim, diagnose the error, and plan a repair.

LAB CASE: UNKNOWN LIQUID

A student uses an electronic balance and a graduated cylinder. The empty container has a mass of 42.31 g. The container plus liquid has a mass of 67.19 g. The liquid volume is 20.0 mL. The accepted density is 1.25 g/mL.

11. Mass of liquid: 24.88 g

12. Experimental density: 1.24 g/mL

13. Percent error: 0.480% (0.48% acceptable)

14. Claim - Evidence - Reasoning

Claim: Is the experimental result close to the accepted value?

Evidence: Cite at least two numbers from the case.

Reasoning: Explain what percent error tells you and what it does not tell you about precision.

The result is close to the accepted value, but the student should repeat trials before judging precision. Evidence: 1.244 g/mL before final rounding vs. 1.25 g/mL and 0.48% error.

15. Error diagnosis

Another student reads the liquid above eye level, rounds every intermediate value, and omits units. Name two problems and one exact repair action.

Read the bottom of the meniscus at eye level, record one estimated digit, keep units through every step, and repeat measurements.

REPAIR LOG

Item/skill to repair: _____

Next action: _____

Incorrect idea or step: _____

When I will retry: _____

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IMAGE CREDITS

Graduated-cylinder photograph: Darrien / Haltopub, Public Domain, Wikimedia Commons, commons.wikimedia.org/wiki/File:Graduated_cylinder.jpg.

Electronic balance, graduated cylinder, and beaker icons: Servier via Bioicons, CC BY 3.0 Unported.
Bioicons: bioicons.com License: creativecommons.org/licenses/by/3.0/

CONTENT SOURCES

NIST SP 811 - Guide for the Use of the International System of Units (SI)
nist.gov/publications/guide-use-international-system-units-si

American Chemical Society

2025 Guidelines and Recommendations for Teaching Middle and High School Chemistry:
acs.org/education/policies/middle-and-high-school-chemistry.html

Lab and Safety Equipment: acs.org/education/policies/middle-and-high-school-chemistry/classroom-and-lab-facilities/safety-equipment.html

The Importance of the Laboratory Experience: acs.org/education/policies/middle-and-high-school-chemistry/teaching-and-assessment/lab-experience.html

Density of Water: acs.org/middleschoolchemistry/lessonplans/chapter3/lesson2.html

Next Generation Science Standards practice - Using Mathematics and Computational Thinking
nextgenscience.org/practice/using-mathematics-and-computational-thinking

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